Auctionmoics-MINI TASK

Mathematics Club- CFI

§1 Introduction

A type of sale where the thing is sold to the highest bidder paintings, players, animals, cars

What do you understand by auction? Give a few examples of places where auctions have taken place. Who was the most expensive player in the **IPL** 2024 auction? How would you justify if they were overpaid, underpaid, or an optimal buy?

↳ Rishabh Pant

↳ many factors involve such as market prices, demand, rarity, market trends, historical price performances, usefulness

§2 A Simple Game Strategy

Consider two manufacturers (A and B) who are competitors for the same market segment for the same product. Each of them wants to maximize their market share and adopts two strategies. The gain or payoff matrix for A when A adopts strategy i and B adopts strategy j is given by a_{ij}. A 2x2 payoff matrix is shown below:

	B1	B2
A1	3	-2
A2	-1	4

1: $3p-q$ $\quad p=q=1/2$
2: $-2p+4q$
1: $3r-2s$ $\quad r=3/5\ \ s=2/5$
2: $-r+4s$

Table 1: Payoff Matrix from A's perspective

During a given time period T, both A and B have to mix strategies. If A plays only strategy 1, then B would play strategy 2 to gain, which A would not want. Each therefore wants to mix their strategies to maximize their gain (or minimize their loss).

§2.1 Questions

1. Write the linear equations governing A's strategy. What is this strategy called? Also, write all the relevant constraints restricting the domain of the variables you have chosen.
   ↳ $3p-q$, $-2p+4q$ ↳ mixed strategy → $p+q=1$
2. Write the linear equations governing B's strategy. What is this strategy called? Also, write all the relevant constraints restricting the domain of the variables you have chosen.
   ↳ $3r-2s$, $-r+4s$ ↳ mixed strategy → $r+s=1$
3. What you have written above are called **Linear Programming Formulations**. Read about the Dual of a Linear Programming Problem and explain how "B's constraints are indeed the Dual problem of A".

primal problem (A's view): A maximizes his minimum expected payoff (setting $p \& q$).
Dual problem (B's view): B minimizes A's maximum expected payoff (setting $r \& s$)

§3 Bidventures

A's constraints leads to equations that define A's optimal strategy & vice versa.

Let's get to know one of the most widely used auction formats:

A **Vickery Auction** (second-price sealed-bid auction) is a common auction format where:

- Bidders submit their bids privately, without knowing the bids of others.
- The highest bidder wins but pays the second-highest bid.
- If there are n bidders indexed as $i = 1, 2, 3, \ldots, n$, each bidder has a private evaluation v_i for the item.
- Bidders submit bid b_i, where b_i is based on their strategy.
- The goal of each bidder is to obtain the maximum utility.

- Maximum Utility:

$$U_i = \begin{cases} v_i - b_{\text{second-highest}}, & \text{if } b_i \text{ is the highest bid,} \\ 0, & \text{otherwise.} \end{cases}$$

§3.1 Let's Understand

1. You have Rs. 50 to spend at PrimeMart. You can buy:
   - A chocolate bar for Rs. 20 each, which you enjoy a lot.
   - Bubblegum for Rs. 10 each, which you like but not as much.

   If your happiness from each chocolate bar is 5 points and your happiness from each bubblegum is 2 points, how should you spend your Rs. 50 to maximize your happiness? (Here, think of **utility** as the happiness points you get from each item.)

   $20x+10y = 50$; $5x+2y \rightarrow$ maximized ; x ; y ; 2 chocolate bars and 1 bubblegum

2. Suppose that there are 3 bidders in a second-price sealed-bid auction. The valuation of each bidder v_i (for $i = 1, 2, 3$) is independently drawn from a uniform distribution over the interval $[0, 1]$.
   - What is the dominant strategy for each bidder (overbid/underbid/truthful bid)? Explain why.
   - Consider the case where one bidder, say Bidder 1, bids **strategically** below their true value. What would happen to the expected payoff of Bidder 1? Would they benefit from underbidding or overbidding?
   - If one bidder, say Bidder 1, knows the valuations of the other two bidders and wants to maximize their expected utility, what should their bid be in terms of the other bidders' valuations?

What is the dominant strategy for each bidder (overbid/underbid/truthful bid)? Explain why.

Overbids: might win the auction but also pay more resulting in -ve utility
underbids: mostly won't win the auction even though v_i might be higher than $b_{\text{second-highest}}$
truthful bidding: bidder wins when $v_i > b_{\text{second-highest}}$ and doesn't overpay.

Consider the case where one bidder, say Bidder 1, bids **strategically** below their true value. What would happen to the expected payoff of Bidder 1? Would they benefit from underbidding or overbidding? → between the two I'd say underbidding although both aren't great

Underbidding: → reduces winning chance
1) lose the auction if $b_1 < 2^{nd}$ highest bid (though $v_1 > b_{\text{second-highest}}$). Utility = 0
2) win the auction if $b_1 > 2^{nd}$ highest bid, utility = $v_1 - b_{\text{second-highest}}$

Overbidding: → risks negative utility
1) win the auction if b_1 is highest but $v_1 < b_{\text{second-highest}}$, utility < 0
2) win the auction if $v_1 > b_{\text{second-highest}}$, no impact though as they still pay $b_{\text{second-highest}}$

If one bidder, say Bidder 1, knows the valuations of the other two bidders and wants to maximize their expected utility, what should their bid be in terms of the other bidders' valuations?

$b_1 = v_{\text{second-highest}} + \delta$ ($\delta \rightarrow$ small increment) ; $v_{\text{second-highest}} = \max(v_2, v_3)$

$U_1 = v_1 - v_{\text{second-highest}}$ is maximized (if $v_1 > v_{\text{second-highest}}$)

If v_1 isn't maximum, then all should bid truthfully (avoids -ve utility)